Stanford MEDICINE

CARDIOTHORACIC TRANSPLANTATION SYMPOSIUM

Challenges to Lung transplant in India

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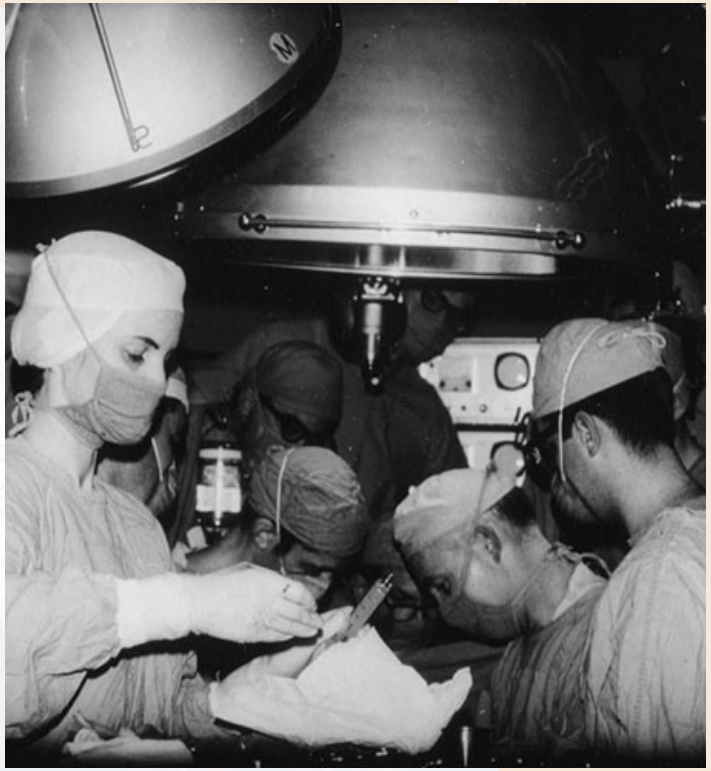
Agenda

- Historical Context of Thoracic Transplantation in India
- Challenges to thoracic organ tx
- Program Initiation and Evolution (2017-2025)
- Trends in Transplant activity and Volume
- Current Scale and Institutional Contribution
- Key Insights

1968 : India’s First Heart Transplant Attempt

- 1968 : First Human Heart Transplant attempt in India Dr P.K Sen
- 1992 AIIMS new delhi – first successful heart transplant in India
- THOTA - transplant act enacted and regulated organ donation ,banned organ commerce. It governs organ transplantation,defines brain death and mandates authorization committees.
- Major amendments were made in 2011/14



A historical timeline

- Thoracic organ transplant in India started in the late 90’s and sputtered in the early decade of this millennium.
- A steady and consistent tick in numbers was a consequence of the rise of cadaveric liver transplant
- early recipients were overseas medical tourists – as indians needed a ‘leap of faith ‘
- As acceptance grew , a rising confidence encouraged established heart surgery programs to test the waters
- These were primarily based in chennai and to an extent Kochi .
- Tamil Nadu and Kerala being prolific with their advocacy with donations after brain death.
- We have started our thoracic organ transplant program in February 2017 in Chennai subsequently Bangalore, Mumbai and Hyderabad

challenges with deceased organ donation

- Organ donation is the primary resource limiting step in a supply demand equation.
- The reasons for slow growth in the national cadaveric programme are diverse & complex

Lack of awareness that voluntary organ donors save lives
Medical unawareness is a significant deterrent
Social, religious beliefs , diverse traditions and taboos in a multicultural society
Absence of a decision-making process in the bereaved family
Slow pace of Sensitisation of doctors on BSD
Bureaucratic logjams

- *In 2019, the organ donation rate in India was a lowly 0.52 per million v/s 36.07 donor per million in the USA.*

….. challenges with deceased organ donation

- We live in two India’s – north versus south
- Currently 1000 per annum organ donors and growing steadily , while the annual RTA incidence of 165000
- We 30 states and 5 union territories – health is a state subject and not under Federal control.
- The wide general acceptance of living related abdominal Tx – spawned a profusion of abdominal organ tx centres
- Thereby all DBD livers and kidneys used locally and never leave city limits .
- This energised thoracic organ Tx
- There are around a few dozen surgical programs that perform an occasional heart transplant (<5 per annum)
- The prolific ones are in (> 25 annually) – ahmedabad, bombay, bangalore, chennai , Hyderabad
- Organ allocation happens on blood group match and date of registration.
- Donor demographics are often incorrect – leading to a very high incidence of size mismatches on evaluation

Donor management obstacles

- Most donor hospitals are in the private sector with homegrown abdominal organ Tx programs
- Anticipatedly – they prioritize abdominal organs over thoracic
- Over hydration and unrestricted fluid administration to drop sodium levels for brain death declaration leads to congested lungs and distended hearts .
- The wanton maladministration of vasoconstrictors and inotropes further harm these organs .
- Any organ not accepted locally is distributed within the state and then nationally by a central body NOTTO
- Abdominal organ Tx teams prefer to operate at night to optimize theater usage by day
- Thus national alerts have a short response time ; this leads to organ wastage, with alerts terminated rather than distributed.
- Opportunity exists in creating a system that maximise utilisation

Challenges at the recipient level

- Late referral due to trust deficit /lack of knowledge/ awareness of therapy ?
- Lack of confidence in outcomes and long term survival
- unrealistic long term expectations of a life post transplant.
- Decompensated and thereby marasmic or obese , poor muscle mass
- Secondary organ dysfunction due to chronic hypoxia/hypercarbia
- The lack of a standardised score (LAS) to justify allocation
- INTERMACS scoring system while a world standard continues to be an esoteric resource with low adaption with cardiologists.
- Physical rehabilitation and dietary supplementation generally not a recommendation and
- lack of local trained rehabilitation therapists and facilities with adequate staffing & equipment

Challenges to HLTx – External environment

- science of organ transplantation is not part of the training curriculum
- Scepticism from prospective patients and referring physicians – feasibility concerns
- Cost and funding constraints – privately funded healthcare
- Perceptibly average, mid to long term survival post Tx – long term outlook
- Our Inability to gauge the true nature and depth of family support.
- Providing long stay rooms that are clean, hygienic, well ventilated in the vicinity are a particular challenge - actively listed waiting patients.
- Managing them long distance, long term frequent biopsies, drug level monitoring, clearance bronchoscopies, airway interventions – a challenge.

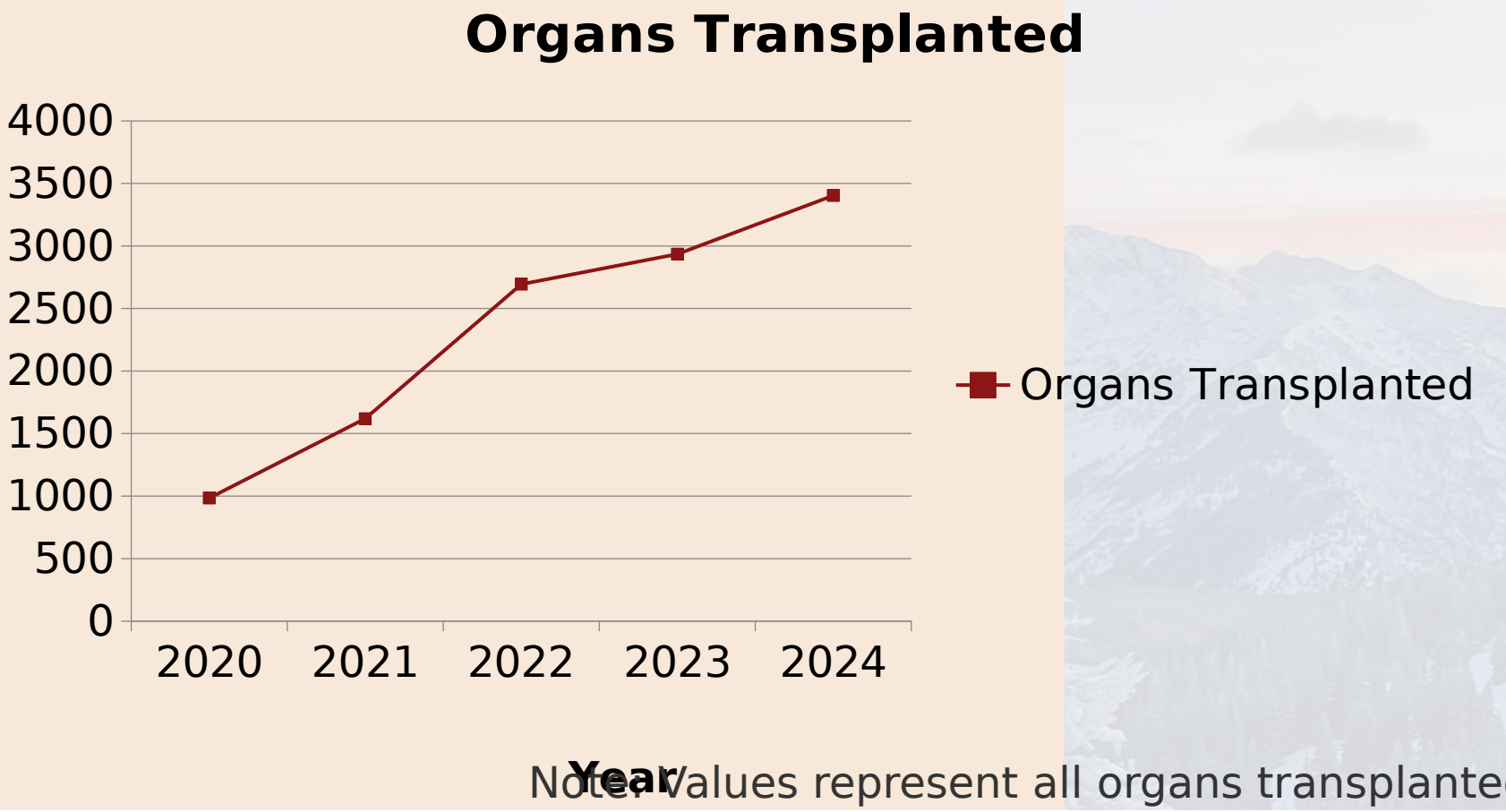
Challenges to an HLTx program - Internal

- Thoracic organ TX is a speciality demands undiluted focus
- Heart teams exist in principle not practise
- Pulmonologists lack exposure and training in transplant immunology – major impediment
- The broad array of instrumentation needed for an HLtx service is a major capital expenditure.
- Pulmonary rehabilitation as part of the respiratory rehabilitation is ill equipped , understaffed and overlooked
- Transplant dietetics and nutrition are a crucial part of care and remain unstructured.
- Immunology and transplant pathology expertise in analysing lung biopsies are growing in recent times.
- A post TX intensive care service is capital and resource intensive, involves a major investment

Program initiation : formation of a dedicated heart – lung transplant team 2017

Following decades of isolated milestones, 2017 marked the establishment of India’s first structured , multidisciplinary heart – lung transplant program

Deceased Donor–Derived Organ Transplants in India (2020–2024)

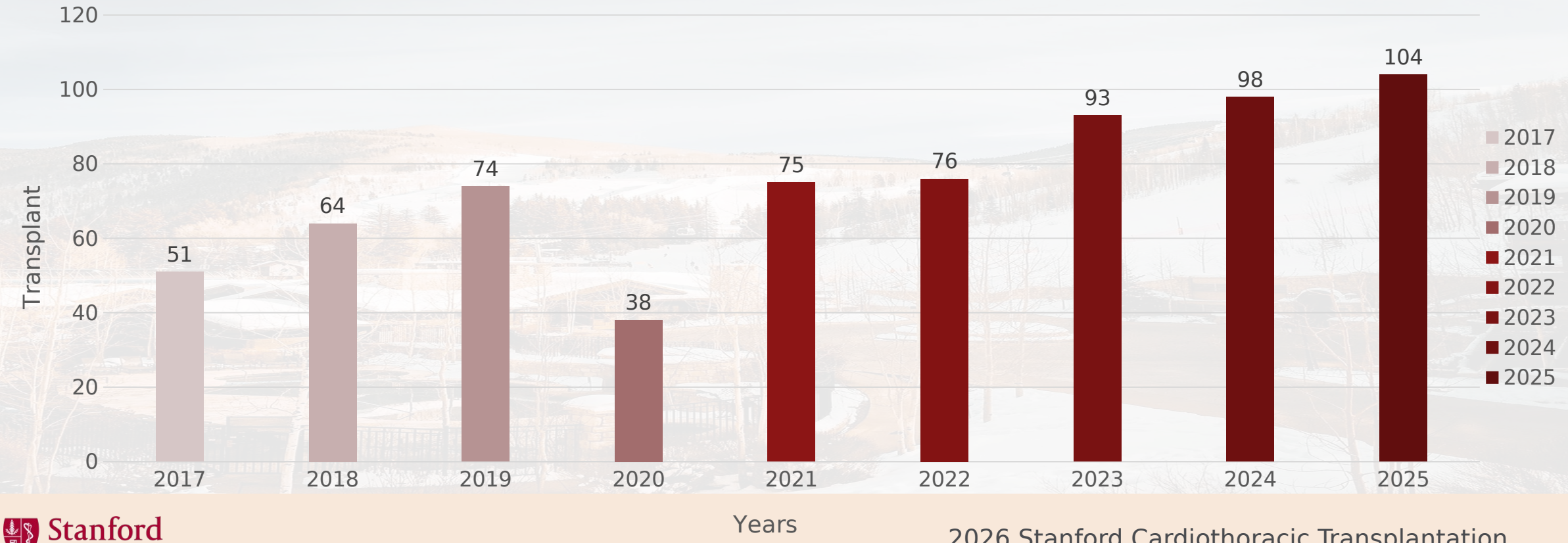


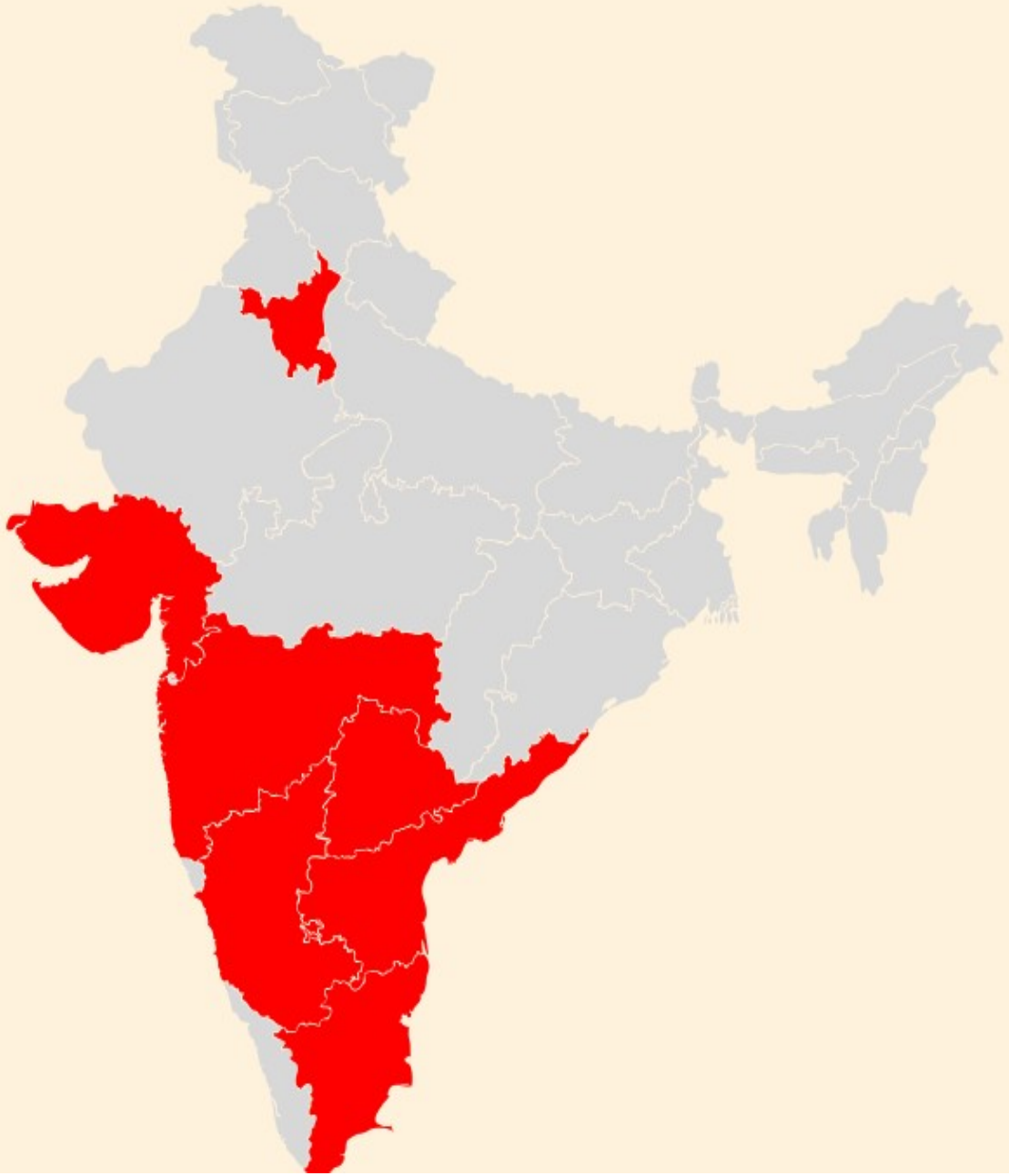
Note: Values represent all organs transplanted, not number of deceased donor

DONOR Contribution

- Southern states have consistently led national deceased organ donation over the last decade in country
- Telangana, Andhra Pradesh , Tamil Nadu and Karnataka form the core donor base

Annual Trends in Heart – Lung Transplant Activity KIMS HLT (2017-2025)





State	Transplant Centres	Number of Transplants
Telangana	KIMS Hyderabad, GHC Hyderabad, Continental Hospital	376
Tamil Nadu	GHC Chennai, RIMC Chennai, Kauvery Hospitals	227
Maharashtra	GHC Mumbai, DY Patil Pune, Sir HN Reliance, Jaslok Hospital	47
Karnataka	Aster Bangalore, KIMS Bangalore, GHC Bangalore	45
Gujarat	KD Hospital, Ahmedabad	17
Delhi (NCR)	Artemis Hospital, New Delhi	2

Total No Of Transplant 714

Mentorship and Capacity Building

From a Single center of excellence to a national ecosystem.

A coordinated, system driven approach to training and mentorship enabling the establishment of heart-lung transplant programs across the country.

Training Pulmonologists , Surgeons across India to build state-level clinical capability

Returning trained specialists to their respective regions to establish regional programs

Decentralizing expertise from the South to Gujarat, Delhi,Maharashtra and other parts of north india.

Vision

The core pillars of scaling thoracic organ transplantation and advancing medical training India

Medical Training
Systemized clinical training to create a sustainable pipeline of transplant physicians and thoracic surgeons.

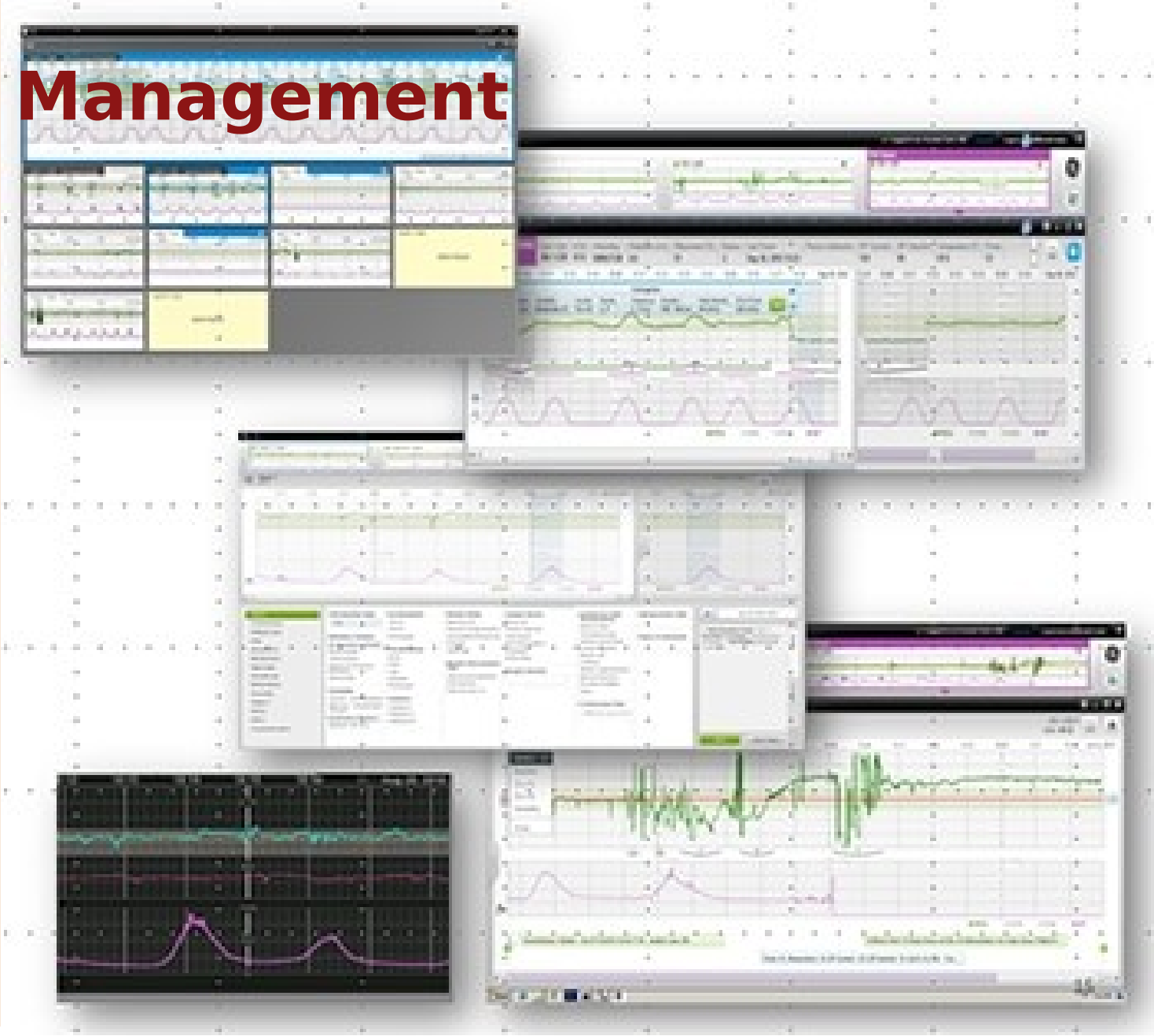
Mentorship
Building regional leadership by empowering local medical Professionals to establish programs.

Systemized Growth
Scaling transplant programs through standardized Protocols, quality frameworks and capacity building.

Global Footprint
Expanding the impact of Indian transplant expertise through international collaboration.

nfrastructure: Practice Management

Centricity
Practice Solution®



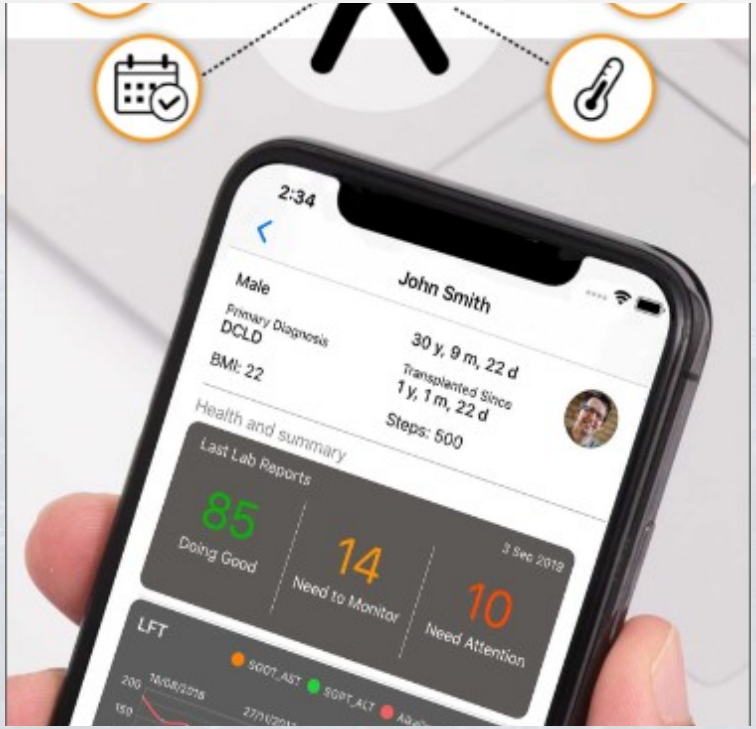
About myTransplant

- My Transplant is a product from Metamagics, a team focused on digital tools for organ transplantation.

- MyTransplant is industry's first and most comprehensive platform that caters to every stage in organ transplantation and caters to every stakeholder, the hospitals, doctors, transplant coordinators, patients and organ allocation organizations.

- There is a web interface, cloud database and dedicated mobile apps for coordinators, doctors and patients.

myTransplant



My Transplant A Complete Platform

Carefully designed for entire organ transplant care ecosystem

Transplant Hospitals

Transplant Surgeons and Clinicians

Transplant Coordinators

Transplant Patients

Transplant Protocols
Organ specific care protocols and treatment guidelines

Workflow Automation
Timely tracking of all care planning over entire transplant lifecycle

Data Harmonization
AI-driven Aggregation and integration of patients' data from multiple sources

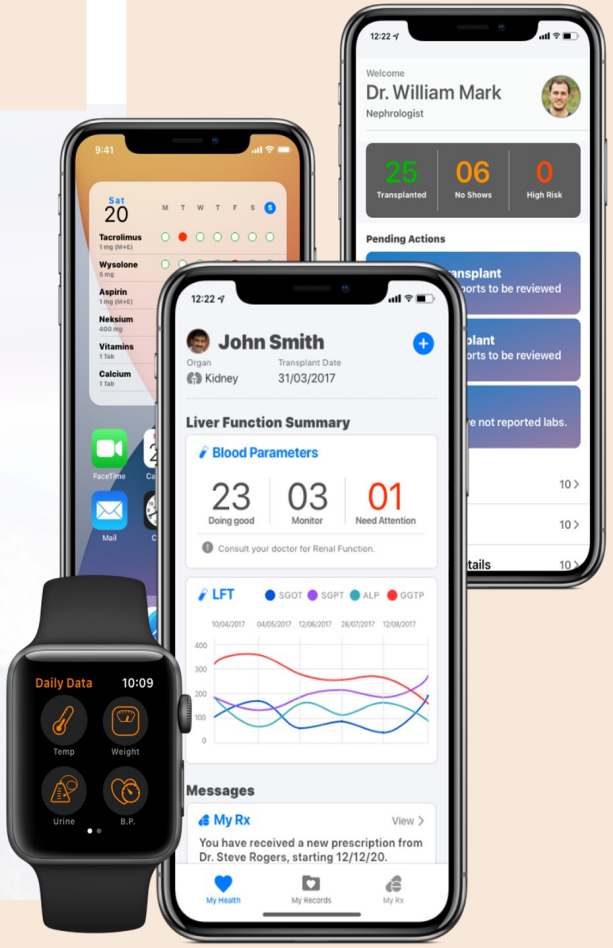
Embedded Analytics
Visual dashboards , insights and intelligence at point of care

Remote Connected Care
In-app video consultation and role specific mobile apps for al stakeholders

Predictive Analytics
Predicting key risk metrics- readmissions, mortality and rejection

Patient Engagement
Tracking and incentivizing adherence

Patient Referrals
Referrals to latest clinical trials , products/services, 3D bio-printed organs



conclusions

India's thoracic organ transplant landscape has evolved from isolated milestones to a national growing ecosystem.

We face persistent challenges in organ donation, regional disparities, out of pocket payer system and poor insurance penetration, insufficient program infrastructure and patient readiness.

Recent progress is driven by structured programs ,capacity buildingand digital innovation with a vision for decentralization and global collaboration

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Thank you

Sandeep attawar

10 Feb 2026